

Special Techniques in Confidentiality (Encryption / Decryption)

Identity-Based Encryption (IBE)

- Removing the need for certificates

Attribute-Based Encryption (ABE)

- encrypt for not a name or identity but a role or attribute (or a collection of attributes)
- Encrypt for a group of developers, people who have been working for 10 yrs
- Set up an access structure : $((A \text{ and } B) \text{ or } (X \text{ and } Y)) \text{ and } (\text{not } Z)$
where A, B, X, Y, Z are attributes or roles.

Fully-Homomorphic Encryption (FHE)

- Can do operations on ciphertext.
↳ searching or computation, sorting...
- Ex: cloud environments (store encrypted data on the cloud).
- Have the cloud service provider do operations on the ciphertext.
- Both query & data are in ciphertext but the operation works

Broadcast Encryption

- Send confidential data to a dynamically-changing user base.
↳ sets of users that change frequently over time
- Ex: Pay TV
- Key Revocation Scheme

Oblivious Transfer

- Alice gives to Bob some data but she does not know which data he actually retrieves
- Bob doesn't see other data

e.g. - Alice sends x_0, x_1 to Bob

- Alice doesn't know which x_i Bob has
- Bob gets x_i but doesn't see x_{1-i}